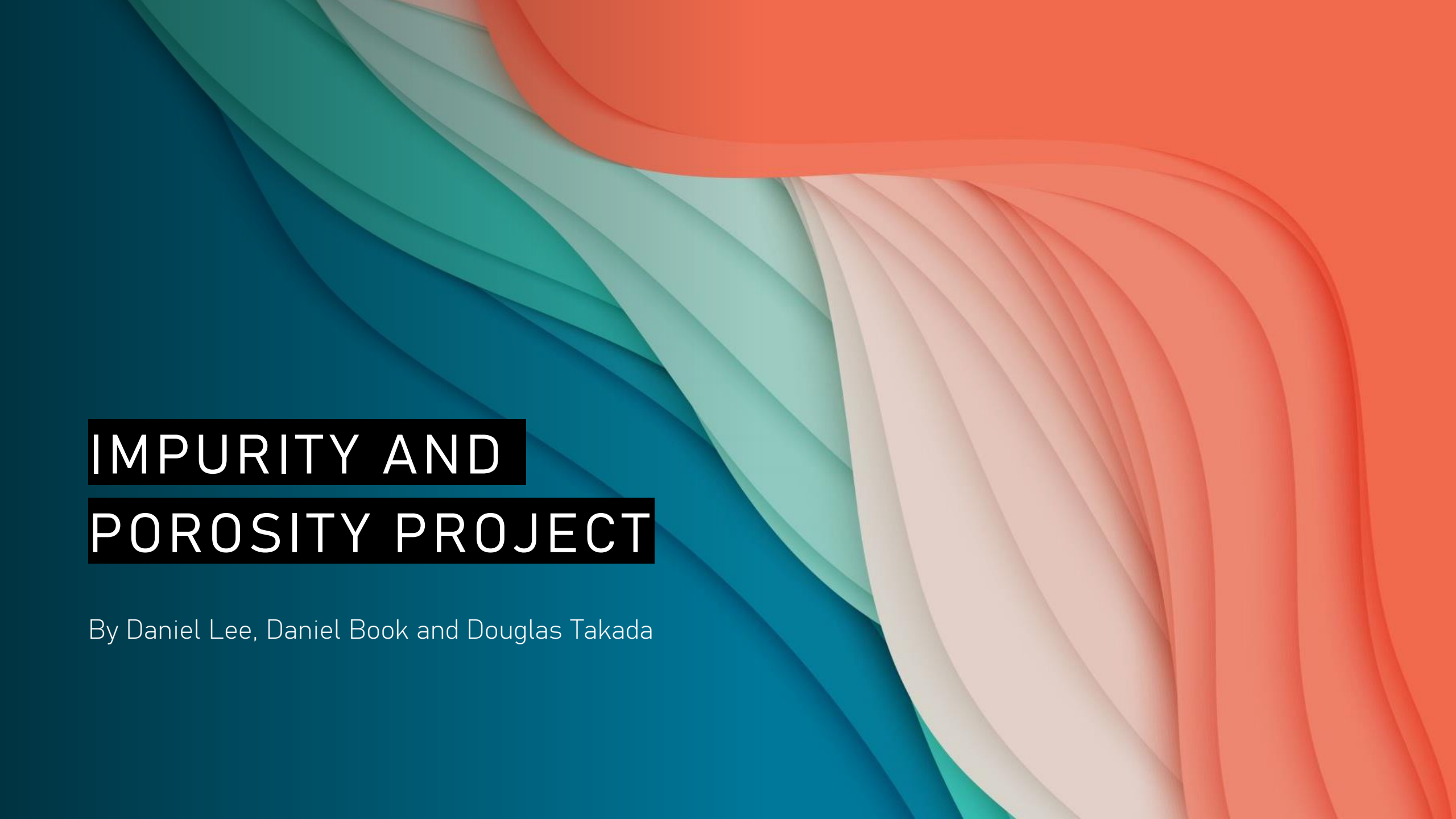




IMPURITY AND POROSITY PROJECT

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ROBOFLOW

- Roboflow has tutorials on how to use various computer vision models.
- All examples run on free GPU-based servers. This limits our image input size, and compiling speed. (Dependent on network connection and availability of GPUs)
 - Amazon SageMaker Studio Lab, Kaggle, Colab (Pay more for more memory allocation)
- Currently looking Meta AI modal Segment Anything Model (SAM) following the Roboflow tutorial from it's Python Notebook file (.pynb) and blog and YouTube video.

SEGMENT ANYTHING MODEL (SAM)

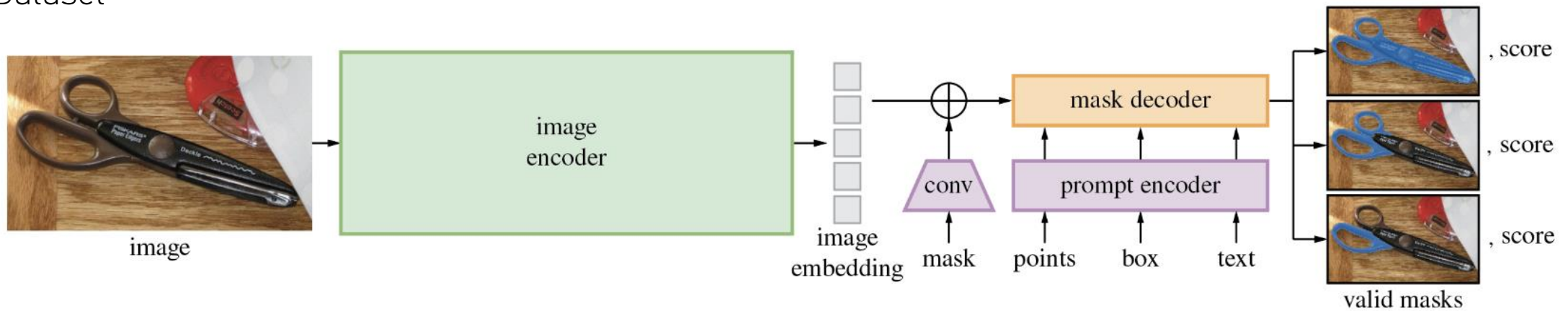
- “The Segment Anything Model (SAM) produces high quality object masks from input prompts such as points or boxes, and it can be used to generate masks for all objects in an image. It has been trained on a **dataset** of 11 million images and 1.1 billion **masks** and has strong **zero-shot performance** on a variety of segmentation tasks.” META AI GitHub
- Masks – Class labels for each pixel – item pixel, item boarder pixel, not item pixel – TensorFlow
- SAM is a instance segmentation model – “Instance segmentation is a deep learning-driven computer vision task that predicts the exact pixel-wise boundaries of each individual object instance in an image.” – IBM
- SAM model checkpoints (encoders) – ViT-B having 91M, ViT-L having 308M, and ViT-H having 636M parameters(slowest) – Roboflow

SAM - CONTINUED

Image -> compresses to a 256x64x64 feature matrix -> Decoder head

Segment model architecture: heavy lifting of image featurization to a transformer model and then trains a lighter model on top. Allowing the smaller model to run featurization with a backend GPU and a web browser

Dataset



SAM – CONTINUED

Using Segment Anything, you can upload an image and:

1. Generate segmentation masks for all objects SAM can identify;
2. Provide points to guide SAM in generating a mask for a specific object in an image, or;
3. Provide a text prompt to retrieve masks that match the prompt (although this feature was not released at the time of writing).

SAM has a wide range of use cases. For instance, you can use SAM:

1. As a zero-shot detection model, paired with an object detection model to assign labels to specific objects;
2. As an annotation assistant, a feature made available in the Roboflow Annotate platform, and;
3. Standalone to extract features from an image, too. For example, you could use SAM to remove backgrounds from images.